

# Project Design Phase

## Proposed Solution Template

|               |                                                  |
|---------------|--------------------------------------------------|
| Date          | 18 June 2026                                     |
| Team ID       |                                                  |
| Project Name  | ShopEz (MERN Full-Stack Application)             |
| Student Name  | JASTI RAGHAVENDRA (jsvpsairaghavendra@gmail.com) |
| Maximum Marks | 2 Marks                                          |

### Proposed Solution Template:

The project team shall provide relevant details within the subsequent solution architecture template.

| S.No. | Parameter                                           | Description                                                                                                                                                                                                                                                                                                                                                                              |
|-------|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.    | <b>Problem Statement<br/>(Problem to be solved)</b> | Contemporary digital retail platforms frequently encounter high latency issues, non-optimized mobile viewing experiences, and convoluted admin workflows. Such technical hurdles result in diminished user engagement and lackluster inventory oversight.                                                                                                                                |
| 2.    | <b>Idea / Solution<br/>description</b>              | ShopEz leverages the MERN ecosystem to provide a high-velocity, single-page application (SPA) experience. The platform integrates live search functionality, persistent shopping cart states, refined category filtering, and a unified management portal for integrated order monitoring and backend control.                                                                           |
| 3.    | <b>Novelty / Uniqueness</b>                         | <ul style="list-style-type: none"><li>• End-to-end JavaScript development using MERN for fluid data synchronization.</li><li>• High-performance media handling via Cloudinary-integrated storage.</li><li>• Adaptive layouts tailored specifically for Men, Women, and Kids categories.</li><li>• Comprehensive administrative access for holistic order lifecycle management.</li></ul> |
| 4.    | <b>Social Impact /<br/>Customer Satisfaction</b>    | Improves the digital shopping experience by offering a secure, intuitive environment. It prioritizes user happiness through a rapid                                                                                                                                                                                                                                                      |

| S.No. | Parameter                                 | Description                                                                                                                                                                                                                                      |
|-------|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |                                           | checkout sequence, immediate stock availability feedback, and transparent history tracking.                                                                                                                                                      |
| 5.    | <b>Business Model<br/>(Revenue Model)</b> | <ul style="list-style-type: none"> <li>• Direct retail sales and platform processing commissions.</li> <li>• Tiered subscription access for professional vendors.</li> <li>• Monetized priority product placement and advertisements.</li> </ul> |
| 6.    | <b>Scalability of the Solution</b>        | The MongoDB NoSQL backend ensures the platform scales horizontally with ease. Planned iterations involve multi-merchant capabilities, AI-based personalized product engines, and expansion into diverse international payment systems.           |